

1. Noether Normalization Theorem

Theorem: Let k be an infinite field and let A be a finitely generated k -algebra. Then there are algebraically independent $y_1, \dots, y_r \in A$ such that A is integrally dependent on $k[y_1, \dots, y_r]$.

Proof: Let $x_1, \dots, x_n$ generate A as a k -algebra. Renumber the x_i such that $x_1, \dots, x_r$ are algebraically independent over k and each of $x_{r+1}, \dots, x_n$ are algebraic over $k(x_1, \dots, x_r)$.

We induct on n . If $n = r$ then A is integral over $k[x_1, \dots, x_r] = A$ trivially. Suppose $n > r$ and the statement holds for $n - 1$ generators. Then x_n is algebraic over $k(x_1, \dots, x_{n-1})$ and there is a nonzero $f \in k(x_1, \dots, x_n)$ such that

$$f(x_1, \dots, x_n) = 0.$$

Let F denote the homogeneous part of largest degree in f .

Lemma: There are $\lambda_1, \dots, \lambda_{n-1} \in k$ such that $F(\lambda_1, \dots, \lambda_{n-1}, 1) \neq 0$.

Proof: For $n = 1$, we get that $F(x) = cx^d$ for some nonzero $c \in k$, the statement follows trivially. Now, for general n , we can view F as

$$F(x_1, \dots, x_n) = \sum_{i=0}^s P_i(x_1, \dots, x_{n-1})x_n^i.$$

Observe that $F(x_1, \dots, x_{n-1}, 1)$ is nonzero since no two P_i have the same degree and there is some nonzero P_i . Indeed, $\deg P_i = s - i$ since F is homogeneous. Since $F(x_1, \dots, x_{n-1}, 1) \in k[x_1, \dots, x_{n-1}]$ is nonzero, there is some $(\lambda_1, \dots, \lambda_{n-1}) \in k^{n-1}$ such that $f(\lambda_1, \dots, \lambda_{n-1}, 1) \neq 0$.

Put $x'_i = x_i - \lambda_i x_n$ for $i = 1, \dots, n - 1$. We show that x_n is integral over $A' = k[x'_1, \dots, x'_{n-1}]$. Observe that

$$f(x'_1 + \lambda_1 x_n, \dots, x'_{n-1} + \lambda_{n-1} x_n, x_n) = 0.$$

The coefficient of the highest power of x_n will be $F(\lambda_1, \dots, \lambda_{n-1}, 1) \in k \setminus \{0\}$; we get this by expanding out F with the same arguments as

in f above. Hence f is monic in x_n which gives x_n is integral over A' . This shows that each $x_i = x'_i + \lambda_i x_n$ is integral over A' being a sum of integral elements.

By the induction hypothesis on A' , there are algebraically independent $y_1, \dots, y_r$ over k such that A' is integral over $k[y_1, \dots, y_n]$. Since an integral extension over an integral extension is still integral, we conclude that A is integral over $k[y_1, \dots, y_n]$. ■